

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2253

Oahe 138 kV
January 31, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2253. The Interconnection Customer proposes a 78.7 MW battery energy storage facility interconnecting at the Oahe 138 kV substation in the Virginia area. The request was received on December 12, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,017,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Canyon Park Substation – Hammerhill 138kV line	Pound – Upper Lake 138kV	110.4	187	13.8
Seaford – Sully Buttes 138kV	Terrora – Quintana 138kV	102.5	265	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Canyon Park – Hammerhill 138kV L	Rebuild line with higher-capacity conductor	\$12,864,000	5% (\$641,000)
Seaford – Sully Buttes 138kV Line	Reconductor line	\$921,000	41% (\$376,000)
	Total allocated to project		\$1,017,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.